

What MongoDB Will Cost BusBee

Storage is not the bill. At 1,000 students you use 0.4% of the free tier — but your 5-second polling loop fires 680 database operations per second, and that is what Atlas charges for.

10 collections analysed bus/api/src/models Atlas pricing · Sep 2026

STORAGE AT 1,000 STUDENTS 1.94 MB Data + indexes, with driver photos. The free tier holds 512 MB.	PEAK OPERATIONS / SECOND 680 500 students watching the map + 50 drivers reporting position.	COST AS BUILT \$57/mo M10 dedicated. Storage would fit free; the op rate does not.
COST AFTER THREE FIXES \$21/mo 250 ops/sec — back inside Flex. Saves \$432 a year.		

SCHEMA

What the database actually stores

Ten collections, defined in `bus/api/src/models` . Before any sizing, one structural fact governs everything below:

THE FINDING THAT DECIDES THE STORAGE ANSWER

Nothing in this schema grows with usage. There is no trip history, no location history, no notification log, no audit trail. `Driver.currentLocation` is a single embedded point *overwritten in place* every five seconds. `stopArrivals` and `notifiedStudentIds` are cleared on every `/trip/start` and `/trip/stop`.

Your database size is therefore a function of **headcount alone**. A bus running this route for three years costs exactly the same storage as one that ran it yesterday.

COLLECTION	ONE ROW PER	WHAT IT HOLDS	ROW SIZE	GROWS WITH
students	rider	name, roll, gender, DOB, address, mobile, bus + stop, OTP, device tokens	515 B	enrolment
drivers	driver	name, DOB, licence, Aadhaar, mobile, address, photo, live position, current issue	650 B / 21 KB	fleet
buses	vehicle	number, plate, capacity, driver ref, route, stops[], notice	1.8 KB	fleet
colleges	campus	name, address, code, planned counts, approval state	350 B	customers
admins	account	adminId, name, gender, DOB, mobile, email, approval, suspension	600 B	customers
staff	console user	name, mobile, role ref, active flag, last login	400 B	customers
roles	permission set	name, colour, module→actions grants, landing page	450 B	customers
banners	singleton	base64 poster image – no size cap in the schema	0–7 MB	–
superadmins	singleton	email, bcrypt hash	150 B	–
counters	sequence	the AD001 admin-id counter	50 B	–

Row sizes are BSON on the wire, uncompressed, computed field-by-field from the schemas. Driver shows without / with a 256 px profile photo.

Where the bytes actually go

Three documents dominate the count. In each, a single field carries most of the weight – and in two of the three, it is not the field you would guess.

Student

One rider. 1,000 of these.

_id · ObjectId	17
college · ObjectId	21
name · String	29
rollNumber · String	27
gender · String	19
dob · Date	13
address · String	59
mobile · String	23
bus · ObjectId	17
stop · String	31
otp · Null	5
otpExpiresAt · Null	14
fcmTokens · Array	187
createdAt · Date	19
updatedAt · Date	19
__v · Int32	9
document framing	5

total	514 B
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One FCM device token is **36% of the whole record** – bigger than name, roll, gender and DOB combined.

Driver · idle, no photo

One driver. 55 of these.

_id · ObjectId	17
college · ObjectId	21
name · String	27
dob · Date	13
gender · String	19
licenceNumber · String	36
aadharNumber · String	31
mobile · String	23
address · String	59
image · Null	7
otp + expiry	19
tripActive · Bool	13
fcmTokens · Array	187
notifiedStudentIds []	25
stopArrivals []	19
currentLocation · Doc	67
currentIssue · Null	14
timestamps + __v	47
document framing	5

total	649 B
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Add a 256 px photo and this becomes **~21 KB** – the image is 32× the rest of the document.

Bus · 15-stop route	
One vehicle. 50 of these.	
_id · ObjectId	17
college · ObjectId	21
busNumber · String	20
plateNumber · String	28
capacity · Int32	14
driver · ObjectId	20
route · String	42
stops[15] · Array	1512
notice · String	73
timestamps + __v	47
document framing	5
total	1799 B
The embedded stops[] array is 84% of the document – roughly 100 B per stop (name, lat, lng, suspended flag, replacement).	

SCENARIOS

Storage, from pilot to scale ceiling

Five scenarios, from a single-route pilot to a hundred thousand riders. Each figure is data plus indexes, uncompressed. WiredTiger's snappy compression takes text down by roughly 3× on disk; base64 photos barely compress at all, so they are the honest ceiling.

SCENARIO	STUDENTS	BUSES	DRIVERS	NO PHOTOS	WITH PHOTOS	OF FREE TIER
A · Pilot, one route cluster	200	10	12	181 KB	421 KB	0.08%
B · Your target	1,000	50	55	882 KB	1.94 MB	0.38%
C · Large university	5,000	200	220	4.14 MB	8.44 MB	1.6%
D · Multi-college SaaS, 20 campuses	20,000	1,000	1,100	17.2 MB	38.7 MB	7.6%
E · Scale ceiling test	100,000	5,000	5,500	86.1 MB	193.5 MB	37.8%

Free tier = Atlas M0, 512 MB. Per-unit costs used: student 745 B, bus 2,030 B, driver 930 B (+20,480 B with photo), ~5 KB fixed per college. All figures include index overhead.

READ THIS ROW

Every scenario fits in the free tier on storage — including 100,000 students. You could grow 100× past your target and still be at 38% of a plan that costs nothing. Storage will never be the reason you pay MongoDB money. Stop worrying about it.

THE REAL COST

You are billed for operations, not bytes

Atlas meters throughput. The free tier caps at **100 operations per second**; the Flex tier prices in bands up to 500; past that you are on dedicated hardware. So the question that sets your bill is not *how much do we store* but *how many queries per second do we fire*.

The answer is: far more than you would expect, because both hot paths run several queries where they need one.

Hot path 1 — the student map poll

`StudentDashboardScreen.tsx:31` polls `/api/student-auth/bus-location` every 5 seconds. Each single poll costs six queries:

```
REQUIRESTUDENT MIDDLEWARE
1  StudentModel.findById - authenticate the bearer token
2  CollegeModel.findById - resolve the owning admin
3  AdminModel.findById - read one `suspended` boolean

ROUTE HANDLER
4  StudentModel.findById - duplicate of query 1
5  BusModel.findById - route and stops
6  DriverModel.findById - live position
```

6 queries ÷ 5 s = 1.2 ops/sec per student watching the map

Hot path 2 — the driver position write

Worse per head. `DriverDashboardScreen.tsx:106` posts to `/api/driver/trip/location` every 5 seconds, and the handler fires a proximity check on every write:

REQUIRED DRIVER MIDDLEWARE

- 1 `DriverModel.findById` – authenticate
- 2 `CollegeModel.findById` – suspension check
- 3 `AdminModel.findById` – suspension check

ROUTE HANDLER

- 4 `DriverModel.findById` – duplicate of query 1
- 5 `driver.save()` – write the new position

NOTIFY ARRIVING SOON · RUNS ON EVERY WRITE

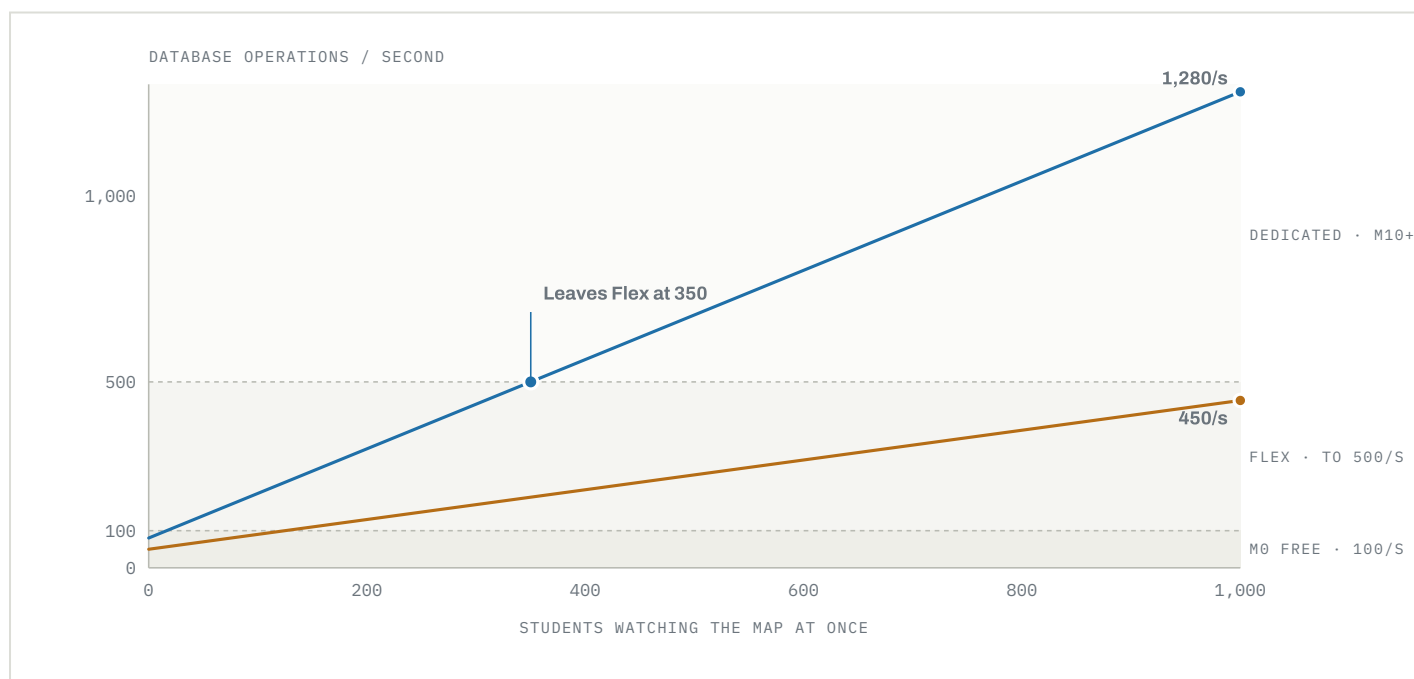
- 6 `BusModel.findOne({driver})` – find the route
- 7 `DriverModel.findById` – third read of the same driver
- 8 `StudentModel.find({...})` – every rider on the bus

8 queries ÷ 5 s = 1.6 ops/sec per driver on a trip

WHAT THIS MEANS AT YOUR SCALE

Fifty drivers on the morning run generate **80 ops/sec** before a single student opens the app. That alone consumes 80% of the free tier's entire budget. Add students and you exceed the free tier at roughly **17 concurrent watchers**.

— As built – 6 queries / 5 s — After the three fixes – 4 queries / 10 s



Both lines include the fixed 50-driver baseline (80 ops/sec as built, 50 after the fixes). Your scenario B peak – around 500 concurrent watchers – sits at 680 ops/sec today and 250 after the fixes.

What each scenario costs, before and after

Assuming half your riders have the app open during the morning run — the realistic peak, since that is exactly when people are waiting at a stop.

SCENARIO	PEAK WATCHERS	OPS/SEC NOW	TIER NOW	COST NOW	OPS/SEC FIXED	TIER FIXED	COST FIXED
A · Pilot	100	139	Flex	\$15	52	M0 free	\$0
B · Your target	500	680	M10	\$57	250	Flex	\$21
C · Large university	2,500	3,320	M30	\$389	1,200	M10	\$57
D · Multi-college SaaS	10,000	13,600	M50	\$1,440	5,000	M30	\$389
E · Scale ceiling	50,000	68,000	M80+	\$5,256	25,000	M60	\$2,844

Atlas list prices, September 2026. M0 free · Flex \$8-30 by ops band · M10 \$57 · M20 \$144 · M30 \$389 · M50 \$1,440 · M60 \$2,844. Dedicated-tier sizing is an estimate — Atlas publishes no ops/sec figure above Flex, so C, D and E need a load test before you commit.

THE LINE THAT MATTERS

At your target the fixes are worth **\$36/month** — **\$432 a year**. One tier up, at a large university, the same three changes are worth **\$332/month**, or **~\$4,000 a year**. The work is a day.

Three changes, in order of payoff

1

Delete the duplicate student lookup

6 → 5 ops

studentAuth.ts:197

 calls

StudentModel.findById

 again for a student the

requireStudent

 middleware already loaded four lines earlier. Attach it to the request and read it. The identical bug is in the driver location handler.

2

Cache the suspension check

5 → 3 ops

Every poll from every user reads `colleges` and then `admins` to fetch one boolean that changes perhaps twice a year. A 60-second in-process cache keyed by college id removes two queries from every authenticated request in the system — not just these two paths.

3

Poll every 10 seconds, not 5

50% of all reads

A bus does not move meaningfully in five seconds, and the GPS fix underneath is not that precise anyway. One constant in `StudentDashboardScreen.tsx:31` . Halves every student-side number on this page.

Together these take the student path from **6 queries every 5 seconds to 3 every 10** — a 4× reduction — and the driver path from 8 to 5. That is the difference between the two lines on the chart.

There is a fourth change worth considering separately, because it is a redesign rather than a fix: **stop polling entirely**. The drivers already push their position to the server every five seconds, so the data is on the server the moment it exists. Students are polling to discover something you could push to them over FCM — which is already wired up and working. That would take the student read load to approximately zero.

RISKS

Three things worth fixing while you are in here

Aadhaar numbers are stored in plaintext

`Driver.aadharNumber` is a required, unique, `/^\d{12}$/` -validated string sitting unencrypted in the driver document. That is an Indian national identity number for 55 people, readable by anyone with database access or a backup file, and it is returned by the drivers list endpoint. Under the Aadhaar Act and the DPDP Act this needs field-level encryption at minimum, and a serious answer to whether you need to collect it at all. Nothing in the app appears to use it.

The banner image has no size cap

Driver photos are capped at 400,000 characters by `parseImageField` in `lib/images.ts:11`. `Banner.imageUrl` has no equivalent — it is bounded only by the 10 MB Express body limit in `app.ts:28`. A single 5 MB poster would occupy more space than all 1,000 student records combined, seven times over. Apply the same cap.

Device tokens accumulate

`fcmTokens` is already the largest field on a student record at 187 B. Sign-in pulls a token off its previous owner, but a user's own stale tokens are only cleaned by `scripts/pruneSharedDeviceTokens.ts`, run by hand. A student who changes phone three times over a degree carries four tokens and a document half again as large. Schedule the prune, or cap the array.

Method. Document sizes are BSON computed field-by-field from the Mongoose schemas in `bus/api/src/models` — type byte, field-name CString, length prefix and payload for each field, plus 5 B of document framing. String lengths assume realistic Indian college data: 18-char names, 45-char addresses, 10-digit mobiles, 20-char stop names, 15 stops per route, one 163-character FCM token. Index overhead is estimated at ~46 B per entry per index across the 21 indexes defined in the schemas.

Operation counts were read directly from the route handlers and middleware — `studentAuth.ts`, `driverTrip.ts`, `lib/suspension.ts` — not estimated. Polling intervals are the constants in the shipped app.

Uncertainty. Peak concurrency of 50% is an assumption, not a measurement; the chart shows the full curve so you can substitute your own. Atlas publishes an ops/sec ceiling only for M0 and Flex, so dedicated-tier placement for scenarios C, D and E is inference from vCPU and RAM and should be load-tested before it becomes a budget line.